Cell Theory & the Plasma Membrane: Revision Paper

Name: …… Class: …… Date: ……

Use this paper to revise before the exam. First read the key points, then answer the practice questions.

Part 1: Key Points

1. History of the cell theory

A **cell** is the basic structural and functional unit of all living organisms.

Cells are so small that they were unknown before the invention of the microscope.

Scientist	Year	Contribution
Robert Hooke	1665	Looked at cork with a simple microscope; saw box-shaped structures he called **cellulae** ("small rooms"). The term "cell" comes from his work.
Anton van Leeuwenhoek	Late 1600s	Designed his own microscope; saw living organisms in pond water, milk and other substances.
Matthias Schleiden	1838	Concluded that all plants are composed of cells.
Theodor Schwann	1839	Reported that animal tissues also consist of individual cells.
Rudolph Virchow	1855	Proposed that all cells are produced from the division of existing cells.

The cell theory (three principles)

1. All living organisms are composed of **one or more cells**.

2. Cells are the **basic unit of structure and organization** of all living organisms.

3. Cells arise **only from previously existing cells**, which pass copies of their genetic material to their daughter cells.

2. Microscope technology

Light (optical) microscope: uses visible light and lenses. **Total magnification = eyepiece × objective** (e.g. 10× × 10× = 100×). Cells are often **stained** because they are tiny, thin and translucent. Visible light limits resolution and scatters, so the maximum useful magnification is about **1,000×**.

Microscope	How it works	Image	Advantages / limits
TEM (transmission electron)	Magnets aim a beam of electrons through thin slices; electrons hit a fluorescent screen. Developed by Ernst Ruska in the 1930s.	Black-and-white shaded image	Up to 500,000×; specimen must be **dead**, sliced very thin and stained with heavy metals.
SEM (scanning electron)	Directs electrons over the **surface** of the specimen.	Three-dimensional image	Only **nonliving** cells and tissues can be observed.
STM (scanning tunneling)	A charged probe tip is brought extremely close to the specimen; electrons "tunnel" across the gap.	3-D computer images of objects as small as atoms	Can be used with **live** specimens.

3. Basic cell types

All cells have a **plasma membrane**, most have genetic material that gives instructions for making needed substances, and cells break down molecules to generate energy.

Organelles are specialized, membrane-bound structures that carry out specific cell functions.

Endosymbiont theory: a symbiotic relationship in which one prokaryotic cell lived inside another, with both benefiting, may have been key to the evolution of eukaryotic cells.

Feature	Prokaryotic cells	Eukaryotic cells
Nucleus	No nucleus	Nucleus containing DNA
Organelles	No membrane-bound organelles	Many membrane-bound organelles
Complexity / size	Simpler and smaller	More complex; generally 1 to 100 times larger
Examples	Bacteria (mostly unicellular); similar to the first organisms on Earth	Multicellular organisms; some unicellular organisms such as yeast
Both have	Plasma membrane	Plasma membrane

4. The plasma membrane
The plasma membrane is a thin, flexible boundary between the cell and its environment. It controls what enters and leaves the cell, and is essential for **homeostasis** (maintaining balance in the internal environment).
A **phospholipid** has a glycerol backbone, two fatty acid chains and a phosphate-containing group. The phosphate makes the **head polar** (attracted to water); the two fatty acid **tails are nonpolar** (repelled by water).
In the **phospholipid bilayer**, two layers of phospholipids are arranged tail-to-tail like a sandwich: heads face the watery environments inside and outside the cell, tails form the interior. Water-soluble substances are stopped by the nonpolar middle.
Fluid mosaic model: phospholipids and proteins move sideways within the membrane like apples floating in water; the different components form a pattern (mosaic) and are in constant motion.
Selective permeability: the membrane allows some substances to pass while keeping others out (like a fish net that lets water through but not fish). How, when and how much enters or leaves depends on the membrane structure.

Component	Location	Function
Phospholipids	Form the bilayer	Form the barrier between inside and outside
Cholesterol	Among the phospholipids	Prevents fatty-acid tails from sticking together; contributes to fluidity
Receptor proteins	Outer surface	Transmit signals to the inside of the cell
Proteins on inner surface	Inner surface	Anchor the membrane to the internal support structure and give the cell its shape
Transport proteins	Span the whole membrane	Create tunnels that move needed substances or wastes across; contribute to selective permeability
Carbohydrates	Attached to proteins, outer surface	Define the cell's characteristics and help cells identify chemical signals

Part 2: Practice Questions

Circle the correct answer (Q1–13)

Question	1	
Which scientist reported in 1839 that animal tissues also consist of individual cells?		
A	Matthias Schleiden	**B**	Robert Hooke
C	Theodor Schwann	**D**	Rudolph Virchow

Question	2	
Which scientist designed his own microscope and observed living organisms in pond water and milk?		
A	Anton van Leeuwenhoek	**B**	Robert Hooke
C	Ernst Ruska	**D**	Theodor Schwann

Question	3	
According to the cell theory, where do new cells come from?		
A	From nonliving matter	**B**	From the nucleus of old cells only
C	From the plasma membrane	**D**	Only from previously existing cells

Question	4	
A light microscope has a 10× eyepiece and a 100× objective lens. What is the total magnification?		
A	110×	**B**	1,000×
C	100×	**D**	10,000×

Question	5	

Why are cells often stained before being viewed with a light microscope?

A	Stain kills the cells so they can be sliced.	**B**	Stain increases the magnification above 1,000×.
C	Stain allows electrons to pass through the cell.	**D**	Cells are tiny, thin and translucent, so stain makes them easier to see.

Question	6	

Which scientist developed the electron microscope in the early 1930s?

A	Anton van Leeuwenhoek	**B**	Ernst Ruska
C	Rudolph Virchow	**D**	Robert Hooke

Question	7	

Why can a TEM not be used to observe living cells?

A	The specimen must be dead, sliced very thin and stained with heavy metals.	**B**	It uses visible light, which kills cells.
C	It only magnifies up to 100×.	**D**	It produces only three-dimensional images.

Question	8	

Which microscope is most suitable for studying the three-dimensional surface of a dead insect?

A	Transmission electron microscope (TEM)	**B**	Scanning tunneling electron microscope (STM)
C	Scanning electron microscope (SEM)	**D**	Simple light microscope

Question	9	

What are organelles?

A	Tiny holes in the plasma membrane	**B**	Specialized membrane-bound structures that carry out specific cell functions
C	Layers of phospholipids	**D**	Substances that surround the cell

Question	10	

Which feature is found in eukaryotic cells but not in prokaryotic cells?

A	A plasma membrane	**B**	Genetic material
C	The ability to break down molecules for energy	**D**	Membrane-bound organelles, including a nucleus

Question	11	

Which arrangement describes the phospholipid bilayer in a watery environment?

A	Two layers arranged head-to-head, with the tails facing the water	**B**	One layer with all tails facing the water
C	Two layers arranged tail-to-tail, with the polar heads facing the water	**D**	Two layers with the polar heads in the middle

Question	12	

A polar, water-soluble substance cannot easily cross the plasma membrane by passing between phospholipids. Why?

A	It is stopped by the nonpolar middle of the bilayer.	**B**	It is attracted to the nonpolar tails.
C	The polar heads are repelled by water.	**D**	Receptors destroy it.

Question	13	

A student says: “The plasma membrane is a fixed, rigid wall.” Which evaluation, based on the fluid mosaic model, is correct?

A	Correct: its components never move.	**B**	Incorrect: its phospholipids and proteins are in constant motion, sliding past one another.
C	Correct: cholesterol locks all molecules in place.	**D**	Incorrect: it contains no proteins.

Questions 14–18: Fill in the blanks using the word bank	
Word bank: bilayer — cell — homeostasis — prokaryotic — selective permeability	
14	The basic structural and functional unit of all living organisms is the _____.
15	Cells that have no nucleus and no membrane-bound organelles are called _____ cells.
16	Two layers of phospholipids arranged tail-to-tail form the phospholipid _____.
17	The property that lets a membrane allow some substances through while keeping others out is called _____.
18	The process of maintaining balance in an organism's internal environment is called _____.

Written Questions

Answer in the space provided (Q19–20)

Question	19	
Compare the TEM and the SEM in terms of:		
(a) how each microscope works **(b)** the type of image each produces **(c)** one limitation of using them		
Answer:		

Question	20	
Describe the structure of the plasma membrane:		
(a) Explain why phospholipids form a bilayer in a watery environment. **(b)** State the role of cholesterol. **(c)** Explain how transport proteins contribute to selective permeability.		
Answer:		

Good luck…